

CS 468 – Secure Programming and Systems

Instructor: Dr. Maha Shamseddine

Project – 3

Topic: Public-Key Infrastructure (PKI) Lab

Collaboration Model: This project can be completed individually or in a team of at most 2 students.

Submission Logistics:

- A **.zip file** : a folder which has all the submission files (refer below) included and later compressed, must be submitted by each member of the team (the same copy of .zip file) to secure an entry in the grade center.
- The .zip file must be named as follows:
P3_[Your Last Name]_[Your GMUID].zip. For instance, if John Smith with GMUID: G12345678 were to submit this file, John would name it: P2_Smith_G12345678.zip
If you are working in a team, the contents of the .zip file must be same, but the name of the .zip file must be changed for each student accordingly.

Submission Files (within .zip file):

1. Detailed lab report (.pdf file) must include:
 - Explanation (what you have done/ observed) for all the tasks.
 - Relevant screenshots (more the better).
 - Contributions made by each group member.
2. A 'readme.txt' file containing names and GIDs of the group members.
3. Include any code files used for completing any of the tasks. (Optional)

NOTE: Submissions that do not meet the requirements given above or simply providing code/ screenshots without any sufficient explanation, will be penalized.

NOTE 2: Please refer the submission guidelines document carefully for formatting and packaging instructions

Expected Results

Task – 1: Start by reading the project documentation.

Include **screenshots** for the commands performed and their outputs, along with detailed description of the steps followed. Show contents of your generated **ca certificate** and **key** using the given commands. Additionally provide your answers for the below questions.

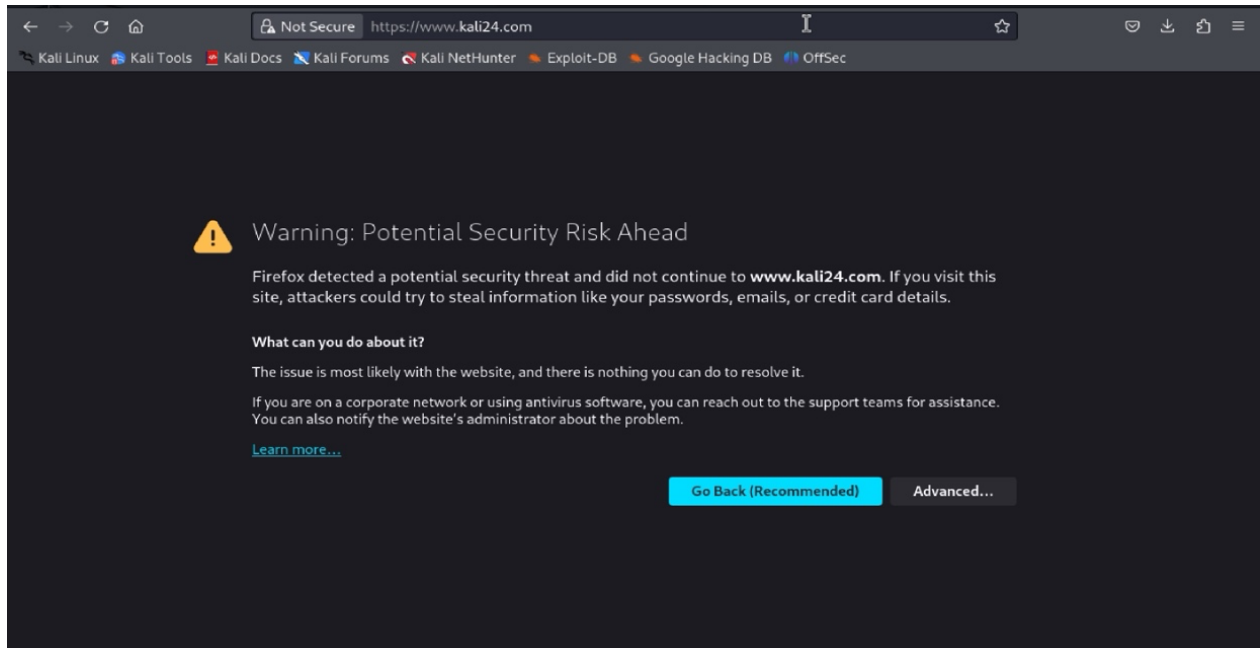
- *Is the certificate part of a certificate chain (e.g., intermediate CA, root CA)? If yes, what is the validity of period (start and end dates) of your certificate? What part indicates this?*
- How long is the key length used in generating the certificate? Which part indicates this?
- *Are there any extensions present in the certificate, and if so, what are they?*
- *In the RSA algorithm, we have a public exponent e , a private exponent d , a modulus n , and two secret numbers p and q , such that $n = pq$. Please identify the values for these elements in your certificate and key files.*

Task – 2: Add **four** alternative names to your certificate signing request. Include **screenshots** for the commands performed and their outputs, along with detailed description of the steps followed. Show contents of your generated **server csr** and **key** using the given commands.

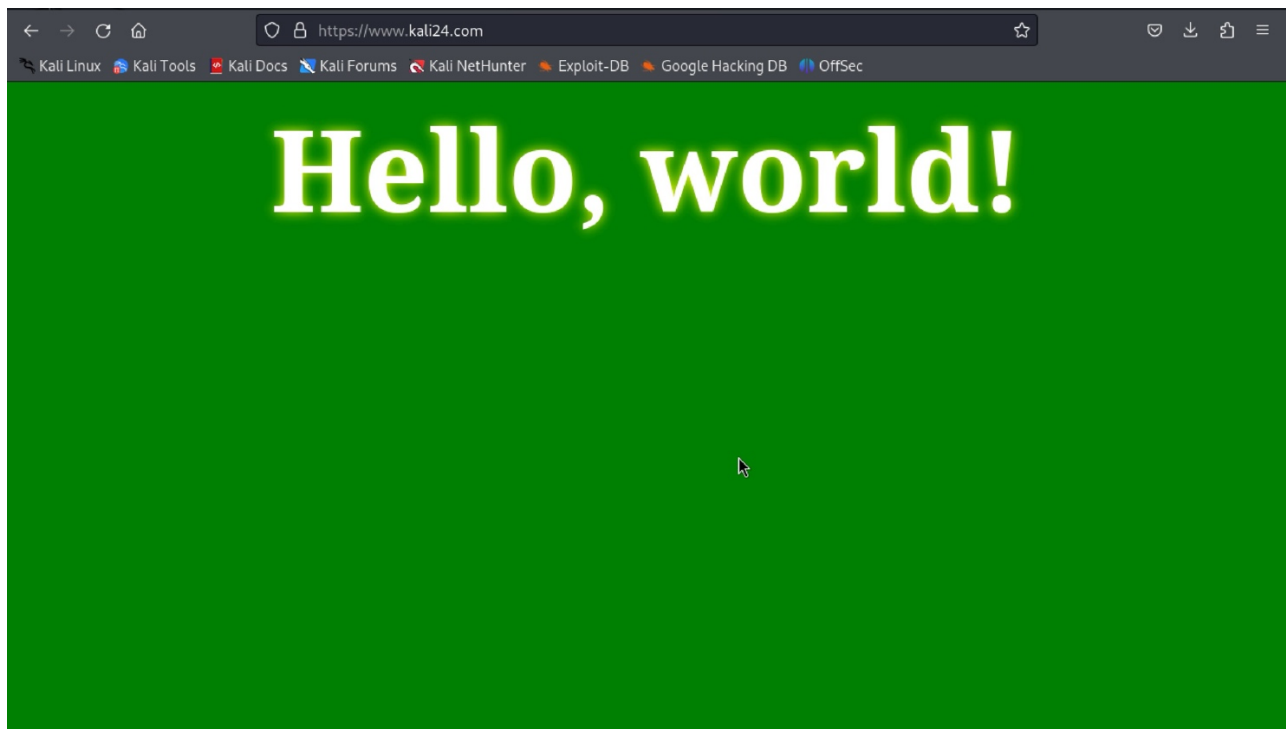
Task – 3: Include **screenshots** for the commands performed and their outputs, along with detailed description of the steps followed. Show contents of your generated **server crt** using the given command.

Task – 4: Include **screenshots** for the commands performed and their outputs, along with detailed description of the steps followed.

When you browse your website initially it should look like this: (Yes! We check your url address)



Once you are able to solve the issue, after you identify it. Your output should look something like:



Please refer the following notes to address common questions

Task-Specific Requirements

1. **All tasks -- written explanations are required.** Screenshots must be accompanied by a description of what you did and what you observed. Submissions with screenshots only, no explanations, will be penalized.
2. **Task 1 -- Answer all five questions in writing** in your report. Pointing to a screenshot of the terminal output is not sufficient; each question requires a written response.
3. **Task 4 -- Docker is mandatory.** Your Apache HTTPS server must run inside the Docker container from the Labsetup folder. Running Apache natively on your VM will not receive credit.
4. **Task 4 --** You must identify and import the correct certificate into Firefox. The spec intentionally does not tell you which certificate to import, you are expected to reason through this yourself and explain your reasoning in the report. Your report must include:
 1. A screenshot of the browser security warning when first visiting your site
 2. A written explanation of why the warning appears and which certificate you chose to import and why
 3. A screenshot showing the trusted HTTPS padlock (no browser warning) as the final outcome
5. **Task 4 -- Your HTML page must display** the course name and the full name and G-number of every team member. Missing this will result in lost credit.
6. **Task 5 (Bonus)-- Explain what you observe when browsing the target website.** Do not skip Step 3. Your report must include a screenshot of the browser's response when visiting the target site and a written explanation of what you observed and why PKI behaves this way.
7. **Task 6 (Bonus) -- Demonstrate a successful MITM attack.** Your report must include a final screenshot showing the browser loads the fake site without raising any security warning. Stopping before this final step will result in a point deduction.
8. **Submission -- ZIP naming:** P3_[LastName]_[GMUID].zip. Both team members submit the same contents, each renamed for their own last name and GMUID.